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COMMERCIAL ITEM DESCRIPTION

TEMPERATURE CALIBRATOR

The General Services Administration has authorized the use of this Commercial Item Description (CID) as a replacement for Class 2 designs of MIL-PRF-28800 for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a Temperature Calibrator that is portable. Its intended purpose is to calibrate for both servo-powered and meter movement (D'arsonval) type temperature indicators. The temperature calibrator will measure and provide simulated source signals for thermocouples, resistance temperature detectors (RTD) and other millivolt transducers. The temperature calibrator will also test synch type temperature indicators. In addition, it will measure thermocouple circuit and thermocouple harness insulation resistance. The temperature calibrator is a support equipment test set that is used for depot and flight line tests.

2. CLASSIFICATION. The temperature calibrator shall be designed in accordance with the Class 2 requirements of MIL-PRF-28800.

3.0 SALIENT CHARACTERISTICS.

3.1 Temperature calibrator. The temperature calibrator shall consist of a high brightness display and adapters to connect and test Movement thermo-couples, meter movement RTDs, servo thermo-couples and commercial thermo-couples. All necessary cables to test each component must be provided with the temperature calibrator.

3.2 Design and construction. The temperature calibrator shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water and sand into critical operating components. It shall be easy to be carried by one person and small enough such that it can be used with ease in small spaces such as a cockpit.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: U. S. AIRFORCE, 404 SCMS/GUEAB, 235 Byron Street Suite 19A, Robins AFB, GA 31098-1670 or SPEC99@us.af.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/ .

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3.2.1 Materials, protective coatings, and finish.

3.2.1.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. However, used, rebuilt, or refurbished items shall not be provided.

3.2.1.2 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Fasteners, handles, and fittings used in the assembly of the item shall also be primed and painted. Surface preparation and pretreatment shall be in accordance with the respective primer and topcoat specifications.

3.2.1.3 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces is acceptable. The use of dissimilar metals only when separated by suitable insulating material is permitted, except in systems where bridging of insulation materials by an electrically conductive fluid can occur. Sealants or gel type gasket materials shall be used between faying surfaces and butt joints.

3.2.1.4 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Painted markings shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible, but shall not be less than ½-inch high. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of SAE-AMS-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of SAE-AMS-STD-595.

3.2.3 Identification and information plates.

3.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the temperature calibrator in a readily accessible location. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer’s name, Commercial and Government Entity (CAGE) code, date of warranty expiration, and National Stock Number (NSN). The temperature calibrator and any components meeting Unique Identification (UID) criteria shall have UID information in accordance with MIL-STD-130 permanently affixed near the respective identification plate(s). UID information shall be included as both a bar code and human readable markings.

3.2.4 Safety.

3.2.4.1 System safety. The design of the temperature calibrator shall not contain any system safety mishap risk categories greater than medium as defined in Table A-IV of MIL-STD-882.

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3.2.4.2 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard.

3.2.4.3 Foreign object damage (FOD). All loose metal parts, such as cable adapters, shall fit neatly in the temperature calibrator's case. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners.

3.2.4.4 Electromagnetic interference (EMI). The temperature calibrator shall be in accordance with the following radiated emission and susceptibility requirements of MIL-STD-461: CE102, CE106, CS101, CS103, CS104, CS105, CS114, CS115, CS116, RE102, and RS103.

3.2.4.5 Electrostatic discharge. The design of the temperature calibrator shall preclude equipment damage due to electrostatic discharge (ESD), protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking.

3.2.5 Human factors. The temperature calibrator shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens and Mission- Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Gear.

3.2.6 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

3.2.7 Welders and welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surfaces to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.2.8 Foolproofness. Where improper installation of an item causes a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.2.9 Service life. The temperature calibrator shall be designed for a minimum service life of five years with no major repairs.

3.3 Environmental conditions.

3.3.1 Operating temperature range. The temperature calibrator shall be capable of operating in ambient temperatures ranging from -40 degrees Celsius to +55 degrees Celsius.

3.3.2 Storage temperature range. The temperature calibrator shall be capable of being stored in ambient temperatures ranging from -40 degrees Celsius to +55 degrees Celsius.

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3.3.3 Solar radiation. The temperature calibrator shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

3.3.4 Salt fog. The temperature calibrator shall be capable of storage and operation in high temperature, high humidity, salt laden, and sea coast environments without damage or deterioration of performance.

3.3.5 Sand and dust. The temperature calibrator shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.

3.4 Weight and dimensions. Overall weight and dimensions shall not exceed: 35 pounds; 20 inches in length; 15 inches in width; 15 inches in height. A handle for carrying the temperature calibrator shall be attached to the case. A mechanism for locking the temperature calibrator shall be provided.

3.5 Transportability.

3.5.1 Surface transportability. The temperature calibrator shall be transportable via all modes of surface shipment: rail, sea, road, etc. and shall be capable to withstand mechanical shock and vibration characteristics of rail, sea, and road transport.

3.5.2 Air transportability. The temperature calibrator shall be transportable for internal air transport in military prime mission cargo aircraft and long range international segment of the Civil Reserve Air Fleet (CRAF). Design criteria can be found in MIL-STD-1791. In all air transport configurations, the temperature calibrator shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The temperature calibrator shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure.

3.6 Maintainability.

3.6.1 Preventive maintenance. The temperature calibrator shall have a recommended preventive maintenance interval (PMI).

3.7 Performance. The temperature calibrator shall be, at minimum, capable of performing diagnostics and calibration.

3.7.1 Malfunction. The temperature calibrator shall be able to simulate individual components therefore identifying which components may be defective.

3.7.2 Power supply. The temperature calibrator shall power DC/AC indicators and simulate temperature input from a thermocouple to the indicators by sourcing: resistance, voltage or current.

3.7.3 Measuring device. The temperature calibrator shall measure voltage, resistance, and current from thermo-couples and millivolt transducers.

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3.7.4 Connection. The temperature calibrator shall connect to all components which might be tested.

3.7.5 Power source. The temperature calibrator shall be powered from an onboard battery provided with the temperature calibrator for up to 3.5 hours or be powered from a wall outlet (115V, 60Hz) without depleting the battery.

3.7.6 Extension cord. The temperature calibrator shall have a 10ft extension from the temperature calibrator to item under test.

3.7.7 Commercial thermocouples. The temperature calibrator shall test commercial type K, T, J, and TF-30 Thermocouples by measuring resistance or voltage.

3.8 Components.

3.8.1 Power cable. The temperature calibrator shall be equipped with an 115V 60Hz power cable which shall plug into a commercial power outlet. A 230V 60 Hz (ac) voltage shall also be included for synchro testing which requires voltage from an aircraft or power supply. The cable must be at least 6ft long and no more than 10ft long.

3.8.2 Resistances check cable. The temperature calibrator shall be equipped with a resistance check cable. The resistance cable must be at least 5ft long and no more than 8ft long. The ends of the cable shall have small alligator type clips.

3.8.3 RTD indicator cables. The temperature calibrator shall have the ability to test all applicable RTD indicators. The RTD indicators shall come with hermetically sealed, electrical receptacle wiring.

3.8.4 Servo and movement thermocouple indicators. The temperature calibrator shall have the ability to test servo and movement thermocouples. The servo and movement indicators shall come with hermetically sealed, electrical receptacle wiring.

3.8.5 Interface. An interface shall be provided. It shall display values, between 50 and 100 degrees F, being measured or sourced with the following accuracy:

a. Temperature values in Degrees Celsius or Fahrenheit, user select +/-1 degree.

b. Resistance +/- .5 Ohm.

c. Voltage (measure or source) +/-.05mV

3.8.6 On/off switch. The temperature calibrator shall have an on/off switch to save power.

3.8.7 Battery indicator. The temperature calibrator shall have a battery charge indicator which shows the state of charge of the onboard battery.

3.8.8 Source/measure. The temperature calibrator shall have a source/measure switch such that it is clear which operation, measuring a voltage or creating a voltage/current level to test a subject, is being performed.

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3.8.8.1 Voltage sourcing capability. When sourcing voltages, the temperature calibrator shall have the ability to tune the output voltage, range 0-35V, down to .5mV of the desired output. The temperature calibrator shall have the ability to source 22.5 or 28.5 +/-.5mV for testing RTD indicators. The temperature calibrator shall have the ability to simulate the voltage range of RTD indicators.

3.8.8.2 External AC voltage. The temperature calibrator shall have the ability to simulate the voltage range of Synchro Indicators using Aircraft AC voltage. The incoming voltage shall be displayed on the indicator. Methods to tune the incoming voltage to the requirements of the item under test shall be provided.

3.8.9 Mode selection switch. The temperature calibrator shall have a mode selection capability such that the level being read or measured can be determined: ohms, current, voltage, AC, DC.

3.8.10 RTD indicator. The temperature calibrator shall have the ability to test single/dual RTD indicators.

3.8.11 Workmanship. The temperature calibrator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs, sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; proper alignment of parts and tightness of assembly fasteners; maintainability and operation of hydraulic lines and fittings.

3.8.12 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be de-burred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.8.13 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be de-burred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the component.

3.8.14 Instructions. Clearly written instructions to use the machine shall be provided on the temperature calibrator's interface.

4. REGULATORY REQUIREMENTS

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

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5. PRODUCT CONFORMANCE

5.1 Product Conformance. The contractor shall certify that the product offered meets the requirements of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product offered for sale in the commercial marketplace.

5.2 Operation and Maintenance Manuals. One copy of the commercial operation and maintenance manual(s), including calibration procedures and adjustments, as applicable, shall be shipped with the equipment.

6. PACKAGING

6.1 Packaging resources. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 7.3). When packaging of materiel is to be performed by Department of Defense (DoD) or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

7. NOTES

7.1 Intended use. The temperature calibrator is intended to be a portable calibrator for both servo-powered and meter movement (D'arsonval) type temperature indicators. The temperature calibrator will measure and provide simulated source signals for thermocouples, resistance temperature detectors (RTD) and other millivolt transducers. The temperature calibrator will also test synch type temperature indicators. In addition, it will measure thermocouple circuit and thermocouple harness insulation resistance. The temperature calibrator is a support equipment test set that is used for depot and flight line tests.

7.2 Source of documents.

7.2.1 The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/.

7.2.2 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://quicksearch.dla.mil or available from the DLA Document Services, 700 Robbins Ave, Bldg. 4, Section D, Philadelphia, PA 19111-5094. Phone (215) 737-8000, or Customer Service DSN 444-8000.

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7.3 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the CID.
b. Packaging requirements (see 6.1)

7.4 RTD indicator cable wiring. The RTD indicators will come hermetically sealed and electrical receptacle wired as follows:

Pin	Connection
A	Power voltage (21-29V DC)
B	Resistance bulb connection
C	Ground and remaining bulb connection
D	5V lighting input (if Applicable)
E	5V lighting ground (if Applicable)

7.5 Servo and movement indicators wiring. The servo and movement indicators will come hermetically sealed and electrical receptacle wired as follows:

Pin	Connection
A	Alumel conductor
B	Open
C	115 V 400-cps input
D	Chromel conductor
E	5V ac or dc input for lighting
F	115V 400-cps and 5V ground. Case ground
G	Open

- see applicable MIL-SPEC for additional information

7.6 National Stock Number (NSNs). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

NSN	6685-01-570-5253

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7.7 Key Words.

Indicators
Meter movement
Portable
Resistance temperature detectors (RTD)
Servo-powered
Support equipment
Temperature measurement
Test set
Thermocouple circuit
Thermocouple harness

7.8 Changes from previous issue. The margins of this CID are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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